The effects of Epigallocatechin-3-gallate on diabetic kidney disease

Presented by Terra Koch

Why this study matters

40% of diabetics experience inflammation resulting in progression to Diabetic Kidney Disease (DKD)

Sterile inflammation is mediated by the NLRP3 inflammasome and is known to have major contributions to DKD

Why this study matters

Inhibiting Thioredoxin-interacting protein (TXNIP) can disrupt TXNIP/NLRP3 interaction→ preventing stimulation of NLRP3 inflammatory complex in podocytes

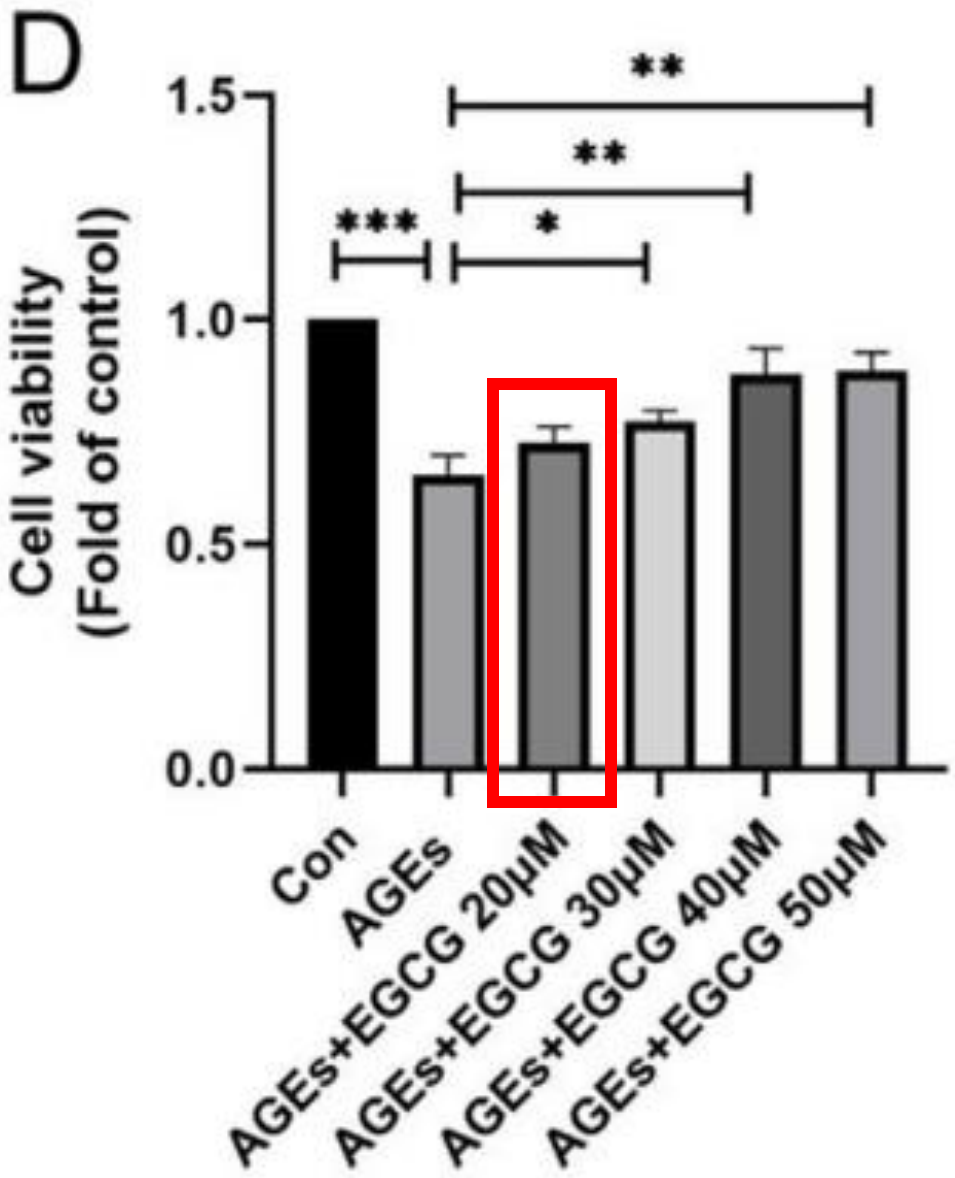
There is significant upregulation of TXNIP and NLRP3 expression in DKD mice and using calycosin and those studies reported to have slowed the progression of DKD by mediating both NLRP3 and TXNIP

Why this study matters

Epigallocatechin-3-gallate (EGCG) is an antioxidant known for anti-inflammatory properties

Studies shows EGCG consumption can reduce inflammation and alleviate oxidative stress in DKD as well as other sterile inflammation-related health conditions and Dapagliflozin (DAPA) is a current treatment for diabetes/kidney disease

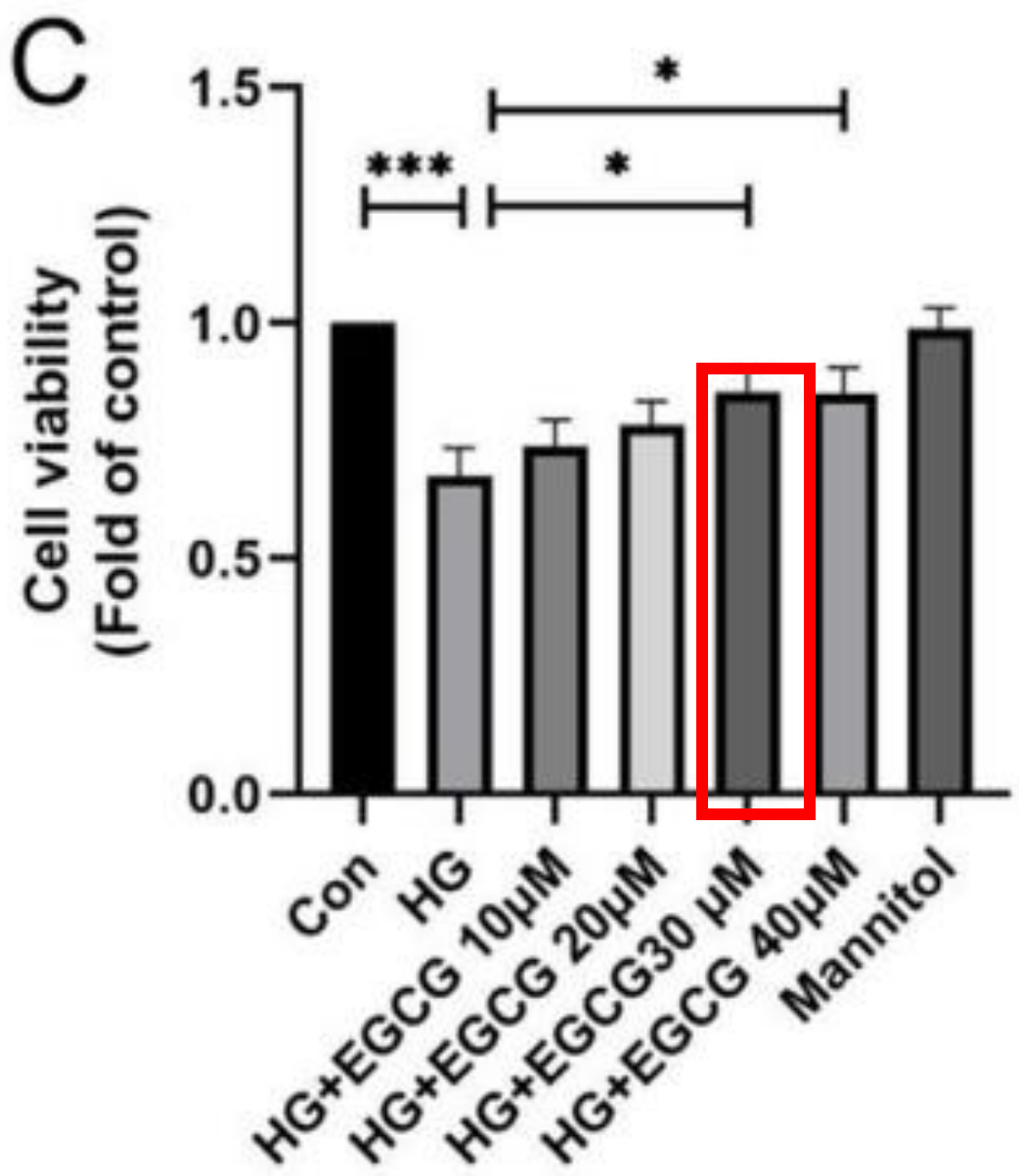
This study evaluates EGCGs impact in human podocytes and DKD mice.



EGCG Increased Podocyte Viability

Podocytes exposed to a high glucose (HG) environment and co-cultured with EGCG and tested for the cell viability, apoptosis, inflammation, and the TXNIP/NLRP3/IL-1β signaling pathway

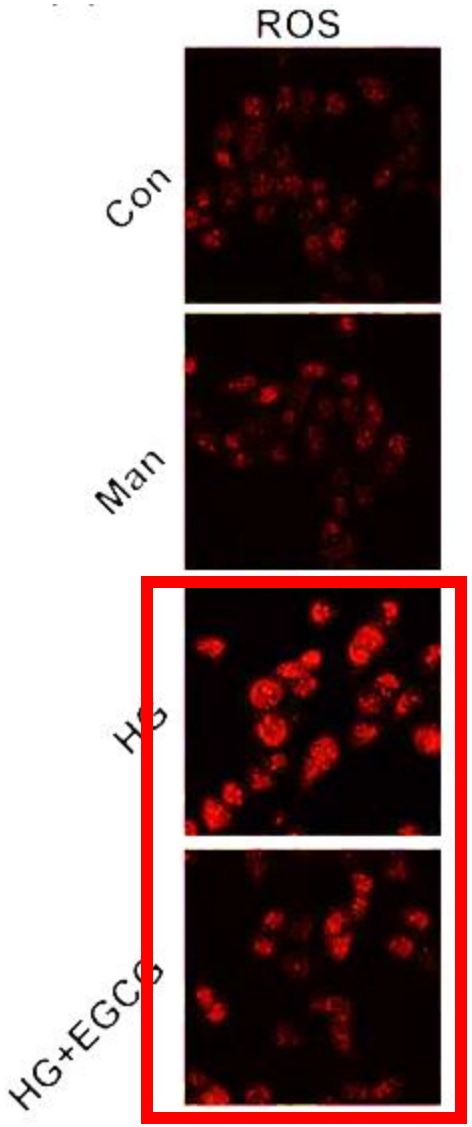
This bar chart shows podocyte viability following exposure to varying concentrations of Advanced Glycation End Products(AGEs)

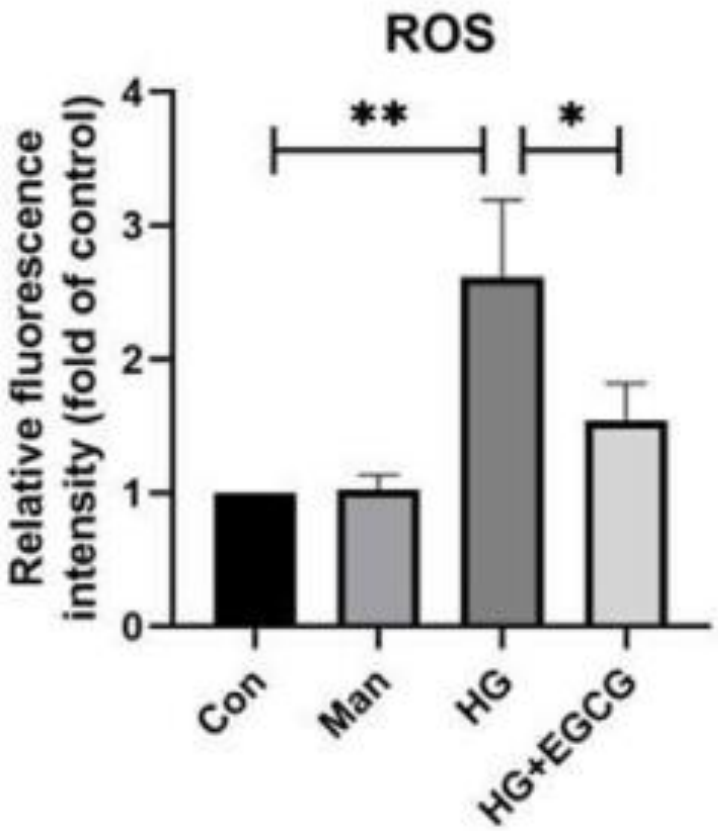


EGCG Increased Podocyte Viability

This bar chart shows podocyte viability following exposure to varying concentrations of High Glucose (HG)

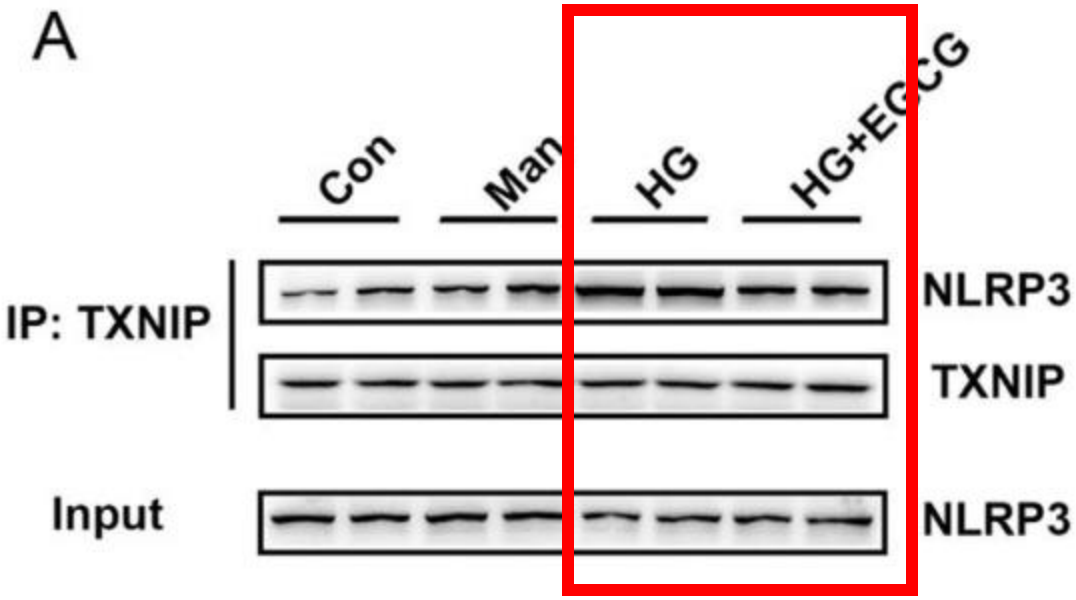
EGCG Decreased Oxidative Stress in Podocytes

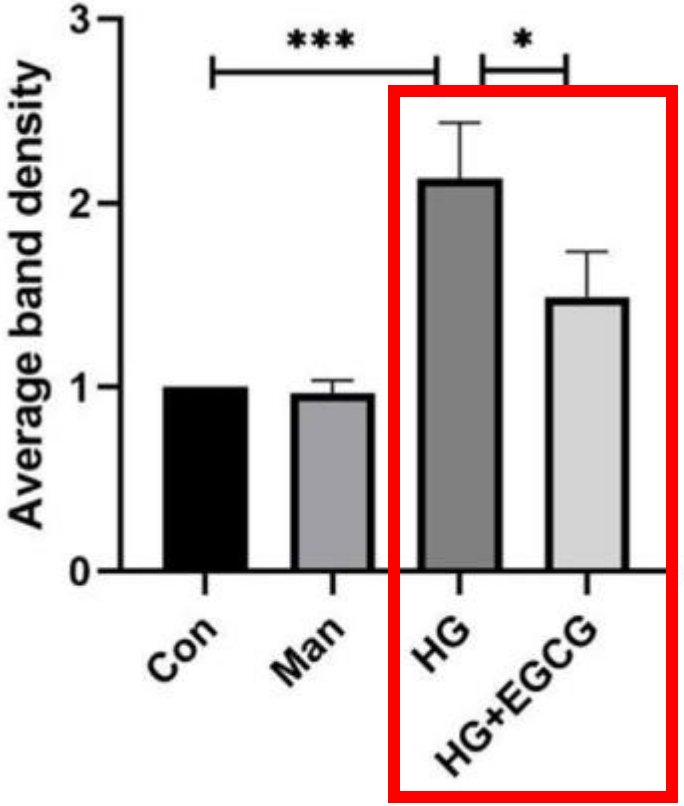




DHE Staining conveying reactive oxygen species (ROS) production in podocytes under varying conditions

EGCG Weakened TXNIP/NLRP3 Interactions

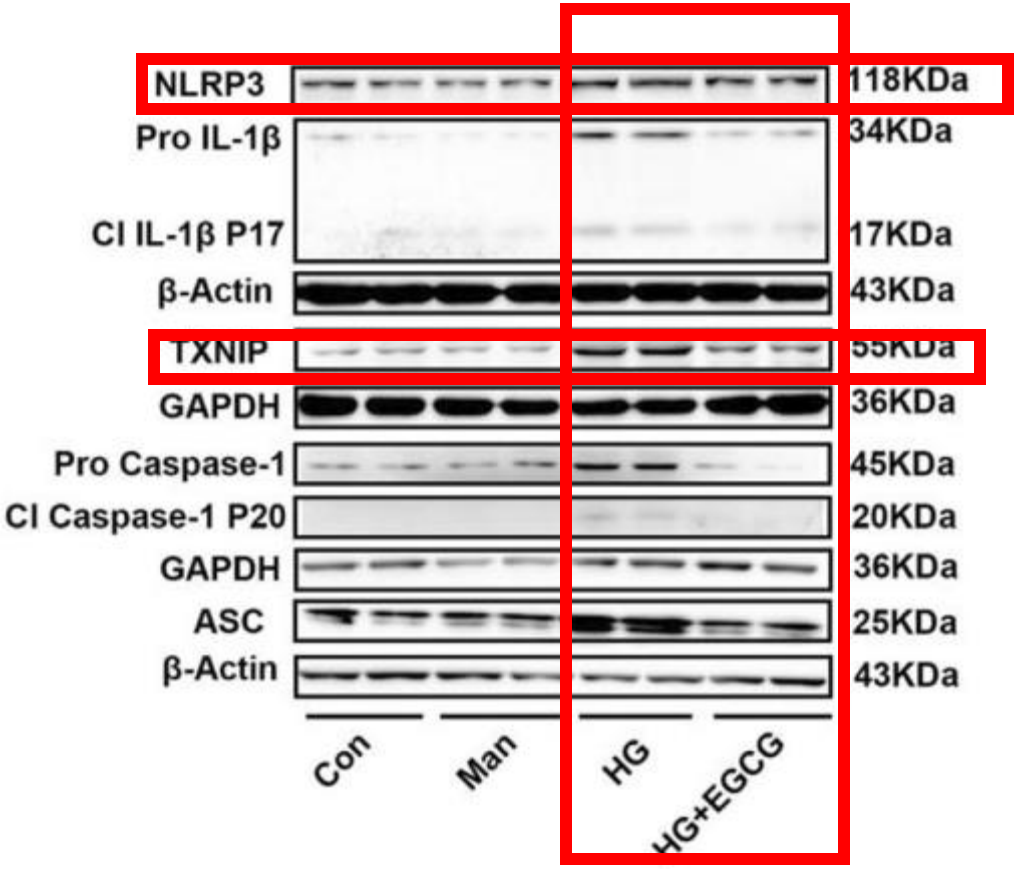




They used western blot to show the relationship between TXNIP and NLRP3 expression when exposed to 30mM of HG and EGCG

NLRP3 and TXNIP antibodies showed significant increases in TXNIP/NLRP3 interaction in HG - treated podocytes

EGCG Suppresses HG-Induced Inflammation in Podocytes



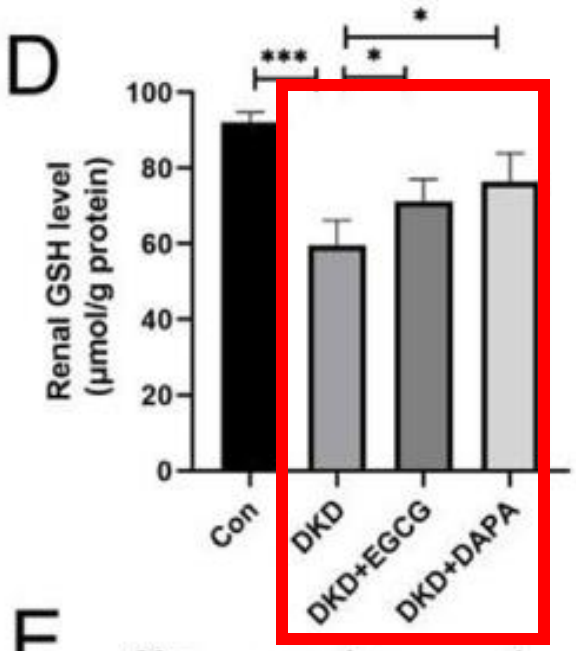
They found EGCG improved kidney function, reduced proteinuria and body weight, and alleviated renal damage.

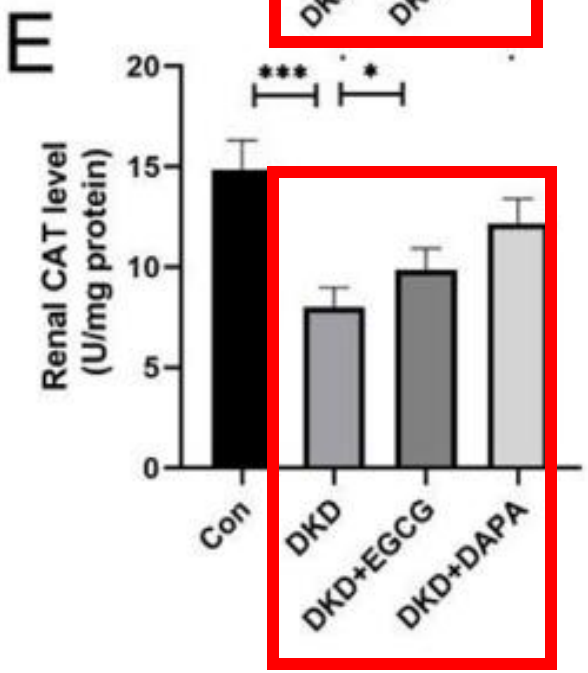
This immunoblot shows EGCG decreased gene and protein expression in TXNIP and NLRP3 EGCG effectively suppresses HG-induced inflammation in podocytes and suppresses NLRP3 inflammasome activation

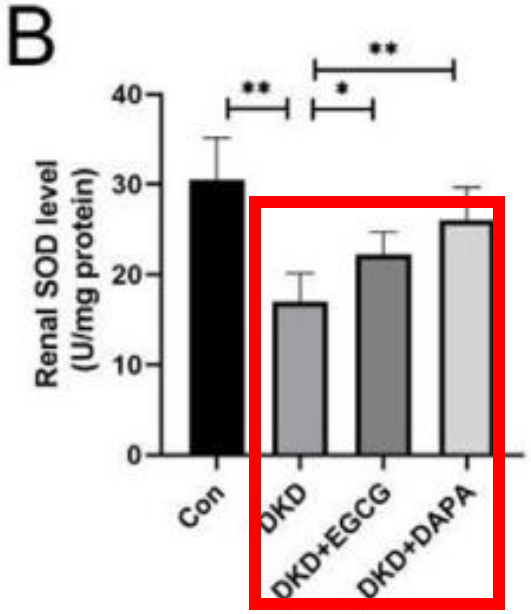
EGCG Alleviates Oxidative Stress and Biochemical Indicators in the Kidneys

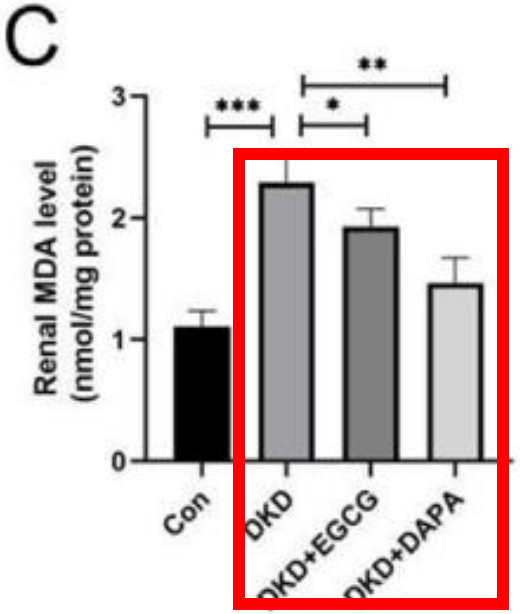
EGCG treated DKD mice exhibit significantly elevated MDA levels, reduced SOD/CAT activities, and declined levels of GSH compared to control group

EGCG treatment especially, with DAPA, significantly elevated GSH, CAT, and SOD levels while declining MDA levels in the kidneys of DKD mice

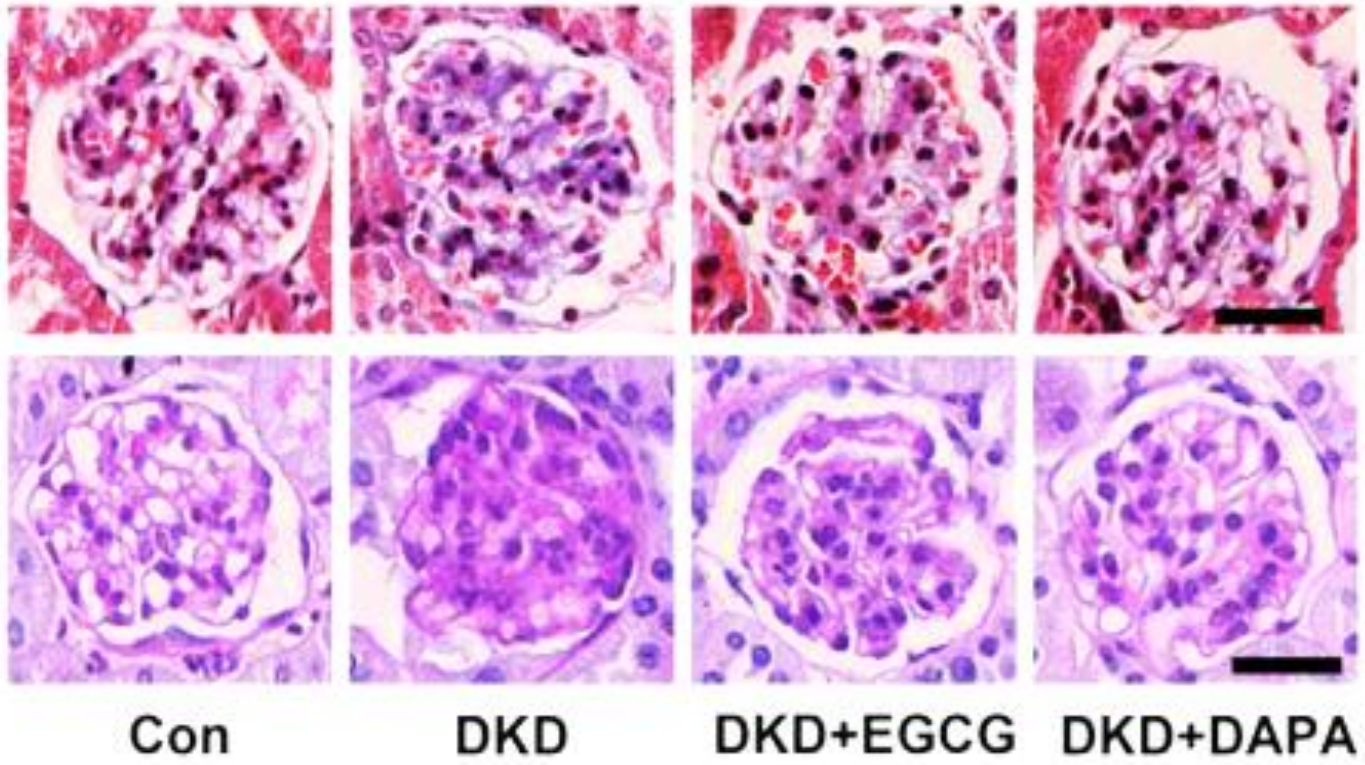






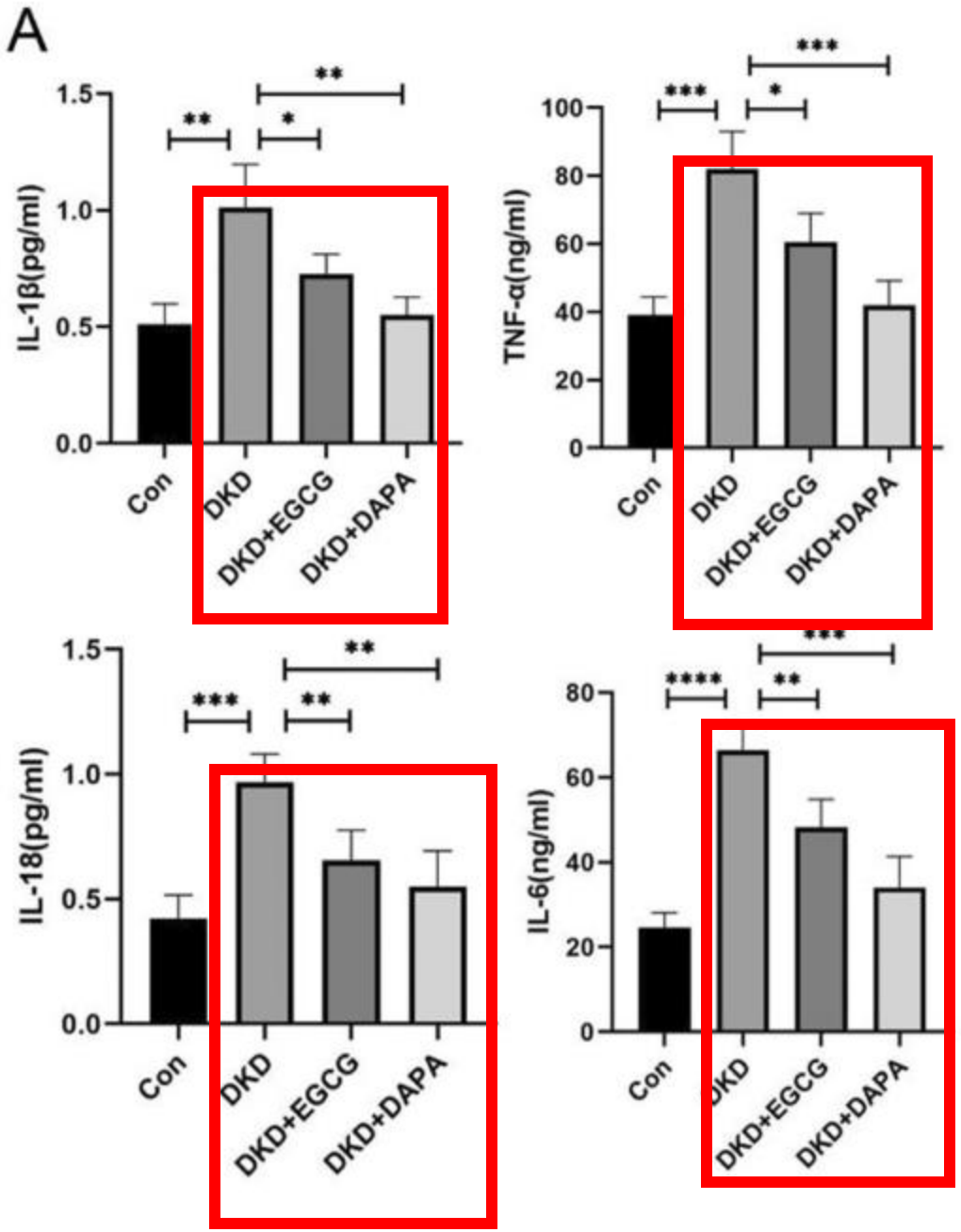


EGCG Alleviated Renal Injury in DKD Mice



They analyzed Kidney tissues with periodic acid-Schiff (PAS) and Masson Staining to examine changes in diseased kidney tissue

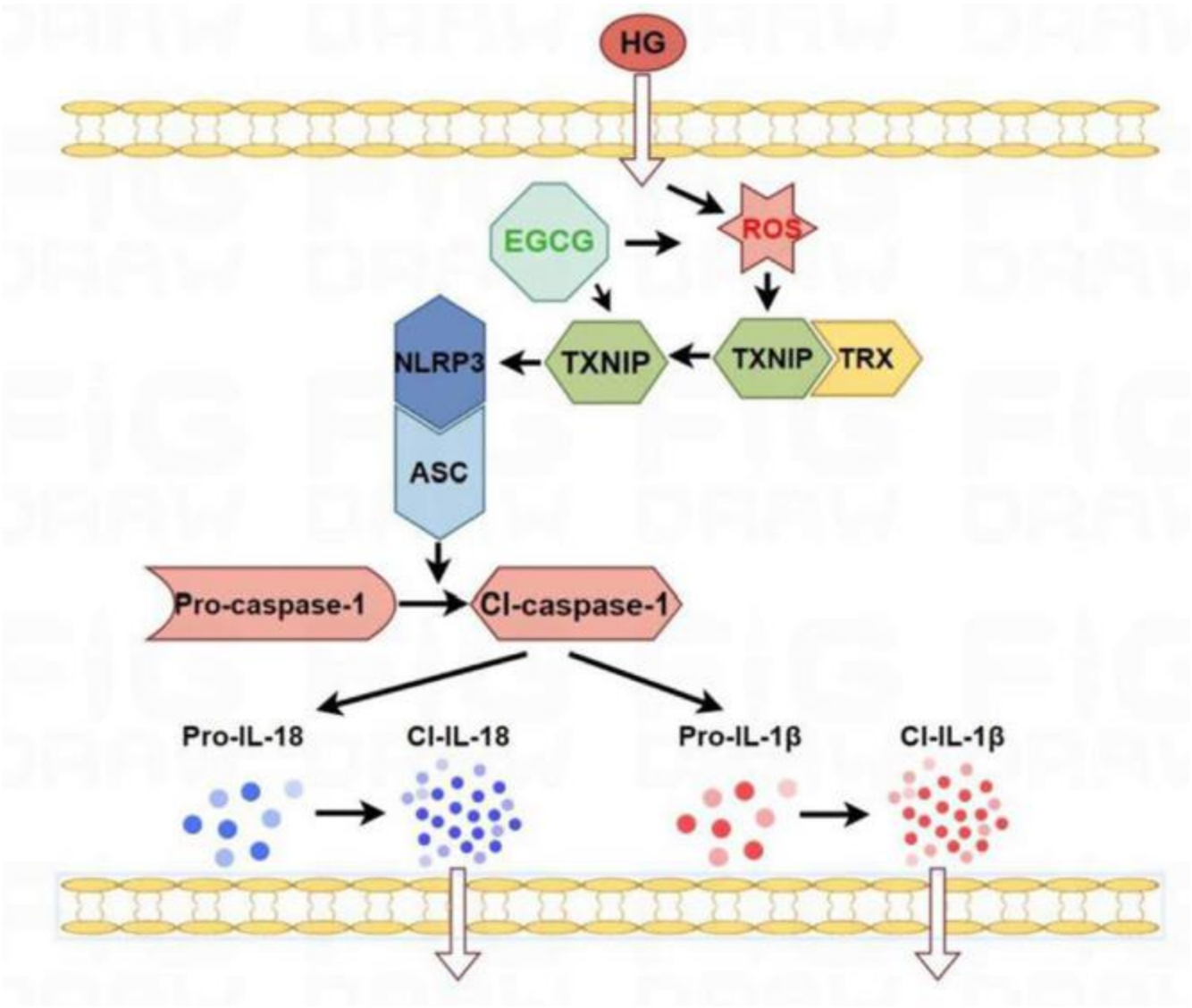
Staining showed that EGCG alleviated renal injury in DKD mice



EGCG Suppresses Inflammation in DKD Mice via TXNIP/NLRP3/IL-1β Axis

EGG treatment decreased levels of inflammatory Factors TNF-α, IL- 6, IL-1β, and IL-18, as well as gene and protein expression of TXNIP and NLRP3 compared to untreated DKD mice

Conclusion



This diagram shows that EGG protects against DKD by reducing inflammation via this pathway

Questions?

Sources

- TXNIP: A key protein in the cellular stress response pathway and a potential therapeutic target - PMC
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